

GAME THEORY & STRATEGY

A Premium University-Level Study Companion

BOOK 2 Quiz 2 Preparation Guide

Textbook + Strategy Handbook + Complete PYQ Solutions
rolled into one self-contained resource

Coverage

Week 5: Repeated Games & Evolution **Week 7:** Matching & Gale–Shapley (ESS)

Week 6: Bayesian, Cooperative & **Week 8:** Cascades, Voting & Arrow
Coopetition

Assessment: Quiz 2 PYQs · Full Solutions · Strategy Guide · Concepts found only in PYQs

How to Use This Book

This book is engineered to be *sufficient on its own* for scoring full marks in Quiz 2. You should be able to put away the slides and lecture videos and study only from here.

- **Part I — Complete Notes (Weeks 5–8).** Every concept is built from first principles. Read the **intuition** boxes first if you are new, then the **definitions**, then the worked examples. No prior economics or mathematics is assumed.
- **Part II — PYQ Solutions.** Every recurring Quiz-2 question is solved in extreme detail (interpretation → topic → thought process → full working → traps → final answer). The official answer keys are reproduced and independently verified.
- **Part III — Concepts Found Only in PYQs.** Tools the exam leans on but the slides state lightly (Bayes mechanics, type-averaged payoffs, discounting/one-shot-deviation, ESS-vs-Nash).
- **Part IV — Quick Revision Sheets.** One-page summary, formula card, algorithm cheat-sheet, trap list, last-minute notes.
- **Part V — Exam Strategy Guide.** How to attack each of the six Quiz-2 families under time pressure.

The box colour code. Definition Intuition / FAQ Why it exists Equilibrium insight
Exam tip / Remember Mistake / Trap / Issue .

★ How marks are won in Quiz 2

Six families generate almost every question across the four papers: (1) an **information cascade** on a 3-1 urn, (2) **voting** (plurality/Borda), (3) **Gale–Shapley** matching, (4) **repeated games** (finite PD + infinite cooperation with discounting), (5) the **Bayesian Battle of Sexes**, and (6) **ESS** in 2×2 games. Master those six templates (Part II) and you have seen the whole exam.

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Part I

Complete Notes (Weeks 5–8)

Week 5 — Repeated Games and Evolution

The two big shifts this week

Quiz 1 lived in *one-shot* games. Week 5 makes two leaps. First, what if the same game is played *again and again*? Repetition lets reputation, punishment and trust enter — and can rescue cooperation from the Prisoner's Dilemma. Second, what if “players” don't reason at all, but strategies spread by *survival of the fittest*? That is evolutionary game theory, whose solution concept is the **evolutionarily stable strategy (ESS)**.

1.1 Repeated Games: the Setup

Stage game, repetition, history

A **stage game** G is an ordinary one-shot game. A **repeated game** plays G many times; each play is a *round* (or *iteration*, *period*). We assume players **observe and remember** the outcome of every past round (*perfect monitoring*). A **strategy** in the repeated game may therefore condition each round's action on the entire *history* of past outcomes — this is the source of all the new behaviour.

Finitely vs. infinitely repeated

Finitely repeated: G is played a fixed, commonly known number of times T . **Infinitely repeated:** play continues forever, *or* ends after each round with some probability (a random, unknown stopping time). The distinction is not pedantic — it completely changes whether cooperation can be sustained.

1.1.1 The stage game: Prisoner's Dilemma

We use the cooperation-friendly normalisation (Cooperate = stay silent, Defect = confess):

		P_2	
		Cooperate	Defect
P_1	Cooperate	2, 2	0, 3
	Defect	3, 0	1, 1

The only stage-game Nash equilibrium is (Defect, Defect) = (1, 1), even though (Cooperate, Cooperate) = (2, 2) is better for both.

1.2 Finitely Repeated Games

Simple Intuition

Play the PD twice. Could the players cooperate in round 1, rewarding each other? Reason backwards: in the *last* round there is no future, so it is just a one-shot PD — both Defect. Knowing round 2 is fixed at (D, D) regardless, round 1 also carries no future consequence, so both Defect there too. Cooperation *unravels* from the end.

Twice-repeated PD: backward induction

Total payoff = sum over the two rounds. Solve the last round first: it is a PD, unique NE (D, D) . Since round 2's outcome is independent of round 1's play, round 1 is also a stand-alone PD $\Rightarrow (D, D)$. **Unique SPNE: defect in both rounds**, payoff $(2, 2)$ total. The same logic gives “always defect” for the PD repeated $T = 1000$ times.

★ The finite-horizon theorem

If the stage game has a *unique* Nash equilibrium, then the finitely repeated game has a *unique* subgame-perfect equilibrium: play that stage NE in *every* round. **If the stage game has *multiple* Nash equilibria**, then richer behaviour — including cooperation in early rounds — can be supported, because a “bad” stage NE in a later round can be used to *punish* deviations earlier.

When early cooperation *can* be sustained (multiple stage NE)

Consider this stage game, played *twice*:

		P_2		
		L	R	C
P_1	U	4, 3	0, 0	1, 4
	D	0, 0	2, 1	0, 0

The stage game has *two* pure NE: $(U, L) = (4, 3)$ and $(D, R) = (2, 1)$. Because there are two, a deviation in round 1 can be punished by switching to the *worse* equilibrium in round 2. Consider the strategies: *Player 1* plays U in round 1, then U in round 2 iff *Player 2* played L in round 1, else D ; *Player 2* plays L in round 1, then the good NE (C/L as appropriate) iff round 1 was (U, L) , else R . The threat “deviate now and we both fall to $(D, R) = (2, 1)$ next round” deters a round-1 deviation, so cooperative-looking play survives in round 1 — a genuine SPNE that gives Nash in *every* subgame. *Moral*: multiplicity of stage equilibria is the door through which cooperation enters finite repetition.

1.3 Infinitely Repeated Games and Discounting

Why This Concept Exists

With infinite repetition you cannot just “add up” payoffs — the sum is infinite for everyone, making comparison meaningless. The fix is **discounting**: a payoff next period is worth less than the same payoff today (impatience, or the chance the game ends). We sum *discounted* payoffs, which converges and ranks strategies cleanly. Crucially, when there is

always a chance of a “next round,” players can use *contingent* strategies (“misbehave now and I punish you later”) — and cooperation can become an equilibrium.

Discount factor and present value

Let $\delta \in (0, 1)$ be the **discount factor**: one unit next period is worth δ today. (If the game survives each period with probability p and payoffs are discounted by r , then $\delta = p \cdot r$.) The present value of a constant stream u forever is the geometric sum

$$u + \delta u + \delta^2 u + \cdots = \frac{u}{1 - \delta}.$$

A higher δ means players are *more patient* / the game is *more likely to continue*.

1.3.1 Trigger strategies

Trigger strategy

A **trigger strategy** begins by cooperating and *keeps* cooperating as long as the opponent does; upon observing a defection it switches (“is triggered”) into a punishment phase of specified length. Two extremes:

- **Grim trigger**: one defection $\Rightarrow$ *punish forever* (revert to D for all future rounds). Longest memory, least forgiving.
- **Tit-for-tat (TFT)**: do to the opponent now whatever they did to you last round. Shortest memory, most forgiving.

1.3.2 When is cooperation sustainable? (Grim trigger)

Deriving the cooperation condition

Suppose both use grim trigger in the PD above ($R=2$ for mutual cooperation, $T=3$ temptation defection, $P=1$ punishment, $S=0$ sucker). Compare, for one player:

$$\underbrace{\text{Cooperate forever} : \frac{2}{1 - \delta}}_{2 \text{ each period}}, \quad \underbrace{\text{Defect now, punished forever} : 3 + \frac{\delta \cdot 1}{1 - \delta}}_{3 \text{ today, then } 1 \text{ each period}}.$$

Cooperation is an equilibrium iff

$$\frac{2}{1 - \delta} \geq 3 + \frac{\delta}{1 - \delta} \iff \frac{2 - \delta}{1 - \delta} \geq 3 \iff 2 - \delta \geq 3 - 3\delta \iff \boxed{\delta \geq \frac{1}{2}}.$$

If players are patient enough ($\delta \geq \frac{1}{2}$ here), **mutual cooperation is sustainable as a subgame-perfect equilibrium**. In general, with temptation T , reward R , punishment

P , cooperation survives iff $\frac{R}{1 - \delta} \geq T + \frac{\delta P}{1 - \delta}$, i.e. $\boxed{\delta \geq \frac{T - R}{T - P}}$ (here $\frac{3-2}{3-1} = \frac{1}{2}$).

The cooperation threshold

$$\delta^* = \frac{T - R}{T - P} \quad (\text{cooperate iff } \delta \geq \delta^*).$$

Bigger *temptation* T raises the bar; bigger *reward* R or harsher *punishment* P lowers it.

1.3.3 Subgame perfection: grim vs. tit-for-tat

Equilibrium Insight

Grim trigger is subgame perfect: after any defection, “everyone defects forever” is itself a Nash equilibrium of the continuation game, so the threat is credible. **Tit-for-tat is generally *not* subgame perfect:** if one player accidentally defects once, TFT prescribes endless *alternating* retaliation $(D, C), (C, D), (D, C), \dots$, which neither player wants to carry out — the punishment is not a continuation equilibrium.

★ Deterrence vs. credibility — the trade-off

Grim trigger: maximal *deterrence* (defection is catastrophic) but the punishment also crushes the punisher, straining *credibility*. **Tit-for-tat:** highly *credible* (proportional, forgiving) but may provide *too little* deterrence. Real trigger strategies (e.g. “punish for k rounds then forgive”) sit between these poles.

Why tit-for-tat is *not* subgame perfect — the arithmetic

Suppose both play TFT and, off the equilibrium path, player 1 has just defected once. TFT now sends them into $\dots, (D, C), (C, D), (D, C), \dots$ — player 1 reaps $3, 0, 3, 0, \dots$ while player 2 suffers $0, 3, 0, 3, \dots$. Player 2’s continuation value from *carrying out* this punishment is $0 + 3\delta + 0\delta^2 + 3\delta^3 + \dots = \frac{3\delta}{1-\delta^2}$, but by simply *forgiving* (returning to cooperation) she would get $\frac{2}{1-\delta}$. For moderate δ forgiving is better, so player 2 would *not* carry out the prescribed retaliation: the punishment is not a Nash equilibrium of the continuation game, hence TFT fails subgame perfection. Grim trigger avoids this because “defect forever” *is* a continuation equilibrium — nobody ever has to do something they’d regret.

A concrete “should I cooperate?” calculation

PD with $R = 2, T = 3, P = 1$ and discount factor $\delta = 0.6$. Cooperation value: $\frac{2}{1-0.6} = \frac{2}{0.4} = 5$. Best one-shot deviation then grim punishment: $3 + \frac{0.6 \cdot 1}{0.4} = 3 + 1.5 = 4.5$. Since $5 > 4.5$, **cooperation is sustainable**. (Consistent with the threshold $\delta^* = \frac{T-R}{T-P} = \frac{1}{2}$: indeed $0.6 > 0.5$.) If instead $\delta = 0.4$: cooperation $\frac{2}{0.6} = 3.33$ vs deviation $3 + \frac{0.4}{0.6} = 3.67$ — now *defection wins*, cooperation collapses.

	Grim trigger	Tit-for-tat
Memory	Infinite (never forgives)	One period (mirrors last move)
Punishment	Defect <i>forever</i>	Defect <i>once</i> , then forgive
Deterrence	Maximal	Moderate
Subgame perfect?	Yes	No (alternating revenge not credible)
Robust to error?	No (one slip = permanent breakdown)	Yes (recovers cooperation)

1.3.4 The Folk Theorem

Feasible and enforceable payoffs

A payoff vector $(u_1, \dots, u_n)$ is **feasible** if some (possibly mixed) combination of stage outcomes yields those average payoffs. It is **enforceable** (individually rational) if each u_i is *at least* the player's *minmax* value — the most they can guarantee themselves — so that deviation can be punished down to that level.

Folk theorem (averaged-utility version)

In an infinitely repeated game, a payoff vector can be supported by some Nash equilibrium *if and only if it is both feasible and enforceable*. *Meaning:* essentially *any* outcome that gives every player at least their guaranteed (minmax) level — including full cooperation — can be an equilibrium when players are patient. Repetition does not pin down a unique outcome; it *explodes* the equilibrium set. (*Example: in the PD, a payoff of 0.5 for player 1 is not enforceable because she can guarantee herself 1 by defecting; 1.2 is enforceable because the opponent can hold her to at most 1.*)

1.4 Evolutionary Game Theory

Game theory without rational players

Standard game theory assumes conscious, rational optimisers. Evolutionary game theory keeps the *mathematics* of games but drops the rationality: strategies are *genetically hard-wired*, and **natural selection replaces deliberate choice**. The question shifts from “what will a rational player do?” to “**which behaviour will persist in a population?**” Survival of the fittest does the optimising.

The dictionary: biology $\leftrightarrow$ game theory

Evolutionary biology	Game theory
Organism	Player
Genetically-fixed behaviour	Strategy
Fitness (reproductive success)	Payoff
Interactions with the population	The game matrix

Fitness is measured *in the context of the whole population*: an organism's success depends on whom it meets, so we average its payoff over random encounters.

1.4.1 The beetles' world (Easley & Kleinberg, Ch. 7)

Beetles compete for food; larger beetles win bigger shares but need more food, so being large is metabolically costly. The “size game” between two beetles:

		Beetle 2	
		Small	Large
Beetle 1	Small	6, 6	1, 10
	Large	10, 1	4, 4

Note this has the *structure of a Prisoner's Dilemma*: Large strictly dominates Small, yet $(\text{Large}, \text{Large}) = (4, 4)$ is worse than $(\text{Small}, \text{Small}) = (6, 6)$.

1.4.2 Evolutionarily stable strategy (ESS)

ESS via invasion

Each organism is repeatedly, randomly paired with others; its *fitness* is its average payoff. Strategy T **invades** S at level x if a small fraction x of the population switches to T (and $1 - x$ keep S). Strategy S is **evolutionarily stable** if there is a threshold y such that for *every* invading T at *any* level $x < y$, the fitness of an S -player *strictly exceeds* that of a T -player. In words: **an ESS cannot be invaded by any small mutant population.**

Is “Small” an ESS? (Invasion by Large at level x)

Population: $1 - x$ Small, x Large. Expected fitness of each type (random pairing):

$$F(\text{Small}) = 6(1 - x) + 1 \cdot x = 6 - 5x, \quad F(\text{Large}) = 10(1 - x) + 4x = 10 - 6x.$$

For small x , $F(\text{Large}) = 10 - 6x > 6 - 5x = F(\text{Small})$ (at $x \rightarrow 0$, $10 > 6$). The mutant Large does *better*, so it spreads: **Small is NOT evolutionarily stable.**

Is “Large” an ESS? (Invasion by Small at level x)

Population: $1 - x$ Large, x Small.

$$F(\text{Large}) = 4(1 - x) + 10x = 4 + 6x, \quad F(\text{Small}) = 1(1 - x) + 6x = 1 + 5x.$$

For small x , $F(\text{Large}) = 4 + 6x > 1 + 5x = F(\text{Small})$. The mutant Small does *worse* and dies out: **Large IS evolutionarily stable.**

The stark lesson

A few large beetles invading a small-beetle world thrive (they rarely meet each other, win most contests); small beetles cannot resist them. The reverse fails. So evolution drives the population to all-Large with fitness 4 — *below* the all-Small fitness of 6. **Evolution can decrease fitness over time:** because each beetle’s “environment” includes all other beetles, the environment grows more hostile as everyone enlarges. This is the evolutionary face of the Prisoner’s Dilemma.

1.4.3 ESS and Nash equilibrium

The relationship, derived

For a symmetric 2×2 game with own-payoffs a (S vs S), b (S vs T), c (T vs S), d (T vs T):

$$(S, S) \text{ is a (symmetric) Nash equilibrium} \iff a \geq c.$$

$$S \text{ is an ESS} \iff a > c, \quad \text{or} \quad (a = c \text{ and } b > d).$$

(The ESS condition comes from requiring $F(S) > F(T)$ under a small invasion: $a(1-x)+bx > c(1-x)+dx$ for small x .)

Equilibrium Insight

Every ESS is a symmetric Nash equilibrium (ESS is the *stricter* concept). The converse fails: if $a = c$ but $b < d$, then (S, S) is a Nash equilibrium yet S is *not* an ESS (a T -mutant does at least as well and then better against itself). ESS = Nash + stability against drift.

The dynamic behind the static test: replicator logic

Why does “higher fitness” translate into “takes over the population”? Because reproduction is proportional to fitness: a type whose fitness exceeds the population *average* grows its share next generation; below-average types shrink. Formally, if p is the share of type S , its growth obeys $\dot{p} \propto p(F(S) - \bar{F})$ (the *replicator equation*). A *rest point* of this dynamic where every nearby perturbation dies out is exactly an ESS. So the invasion test ($F(S) > F(\text{mutant})$ for small mutant share) is the *static shadow* of a dynamic stability condition — no organism optimises; the maths of differential reproduction does it for them.

	Nash equilibrium	ESS
Players	Rational optimisers	Hard-wired; selection optimises
Test	No profitable deviation	No mutant can invade (stricter)
Condition (2×2)	$a \geq c$	$a > c$, or $a = c$ & $b > d$
Logic	Best response	Replicator stability
Relationship		Every ESS is a symmetric NE; not conversely

Memory aid

“**ESS = Nash that survives mutants.**” Check Nash first ($a \geq c$); then ask the extra question — *if a few mutants drift in, do they do strictly worse?* A *strict* Nash ($a > c$) always passes; a *tie* ($a = c$) passes only if you beat the mutant when it meets itself ($b > d$).

1.4.4 Mixed/polymorphic ESS: the Hawk–Dove game

Simple Intuition

Hawk–Dove explains why aggression exists in a population but does not take over completely. Two individuals contest a resource of value V ; fighting (two Hawks) costs the loser C , with $C > V$. The population settles at a *mix* of Hawks and Doves.

		Player 2	
		Hawk	Dove
Player 1	Hawk	$\frac{V-C}{2}, \frac{V-C}{2}$	$V, 0$
	Dove	$0, V$	$\frac{V}{2}, \frac{V}{2}$

Let a fraction p of the population play Hawk. A focal individual’s fitness:

$$F(\text{Hawk}) = p \cdot \frac{V-C}{2} + (1-p)V, \quad F(\text{Dove}) = p \cdot 0 + (1-p)\frac{V}{2}.$$

At a (mixed) ESS the two types must have *equal* fitness (else the fitter spreads):

$$p \frac{V-C}{2} + (1-p)V = (1-p)\frac{V}{2} \implies \boxed{p^* = \frac{V}{C}}.$$

Numerical Hawk–Dove ESS

With $V = 6$, $C = 10$: $F(\text{Hawk}) = 6 - 8p$ and $F(\text{Dove}) = 3 - 3p$. Setting equal: $6 - 8p = 3 - 3p \Rightarrow 3 = 5p \Rightarrow p^* = 0.6$. So 60% Hawks, 40% Doves is the stable mix — matching $p^* = V/C = 6/10$. If Hawks exceed 60%, fighting is too costly and Doves do better (Hawk share falls); below 60%, Hawks find easy prey and spread. The mix self-corrects — a *polymorphic* ESS.

1.4.5 The 1:1 sex ratio as an ESS**Equilibrium Insight**

Why are sex ratios near 1:1? If females were rarer, a newborn female would have better mating prospects and more offspring, so genes biasing toward daughters would spread — until females become common and the advantage vanishes. The symmetric argument holds for males. The only ratio immune to invasion by a “produce-more-of-the-rarer-sex” mutant is 1:1. A classic Fisher/Hamilton ESS argument with no conscious optimisation anywhere.

End of Week 5**Quick Revision — Week 5**

Repeated games: strategies condition on history. *Finite* horizon + unique stage NE $\Rightarrow$ unravelling (always play stage NE); *multiple* stage NE $\Rightarrow$ cooperation possible via punishment. *Infinite* horizon: discount with δ ; PV of constant u is $u/(1 - \delta)$. **Grim trigger** sustains cooperation iff $\delta \geq \frac{T-R}{T-P}$ ($= \frac{1}{2}$ for the standard PD); it is SPE, TFT is not. **Folk theorem:** feasible + enforceable $\Rightarrow$ supportable. **ESS:** a strategy no small mutant can invade; $a > c$ (or $a = c, b > d$); every ESS is a symmetric NE, not conversely. **Hawk–Dove ESS:** $p^* = V/C$.

Formula Sheet — Week 5

$$\sum_{t=0}^{\infty} \delta^t u = \frac{u}{1 - \delta}; \quad \text{cooperate (grim) iff } \delta \geq \frac{T - R}{T - P}.$$

ESS (symmetric 2×2): S is ESS iff $a > c$, or $(a = c \text{ and } b > d)$. **Hawk–Dove:** $p^* = \frac{V}{C}$ (fraction Hawks).

★ Algorithm sheet

Finite repetition: count stage NE $\rightarrow$ if unique, answer “stage NE every round”; if multiple, look for punishment-supported cooperation. **Sustain cooperation?** compute $\delta^* = \frac{T-R}{T-P}$ and compare to given δ . **ESS test:** write $F(S), F(T)$ under invasion at level x ; S is ESS iff $F(S) > F(T)$ for small x . **Mixed ESS:** set $F(\text{type 1}) = F(\text{type 2})$, solve for the population fraction.

Practice questions.

1. *Discounting.* If $T = 4, R = 3, P = 1$, find the minimum δ for grim-trigger cooperation. [$\delta \geq \frac{1}{3}$.]
2. *Finite.* Explain why the PD repeated exactly 100 times has “always defect” as its unique SPNE.

3. *ESS*. In the beetle game, verify Large is ESS and Small is not using the invasion fitnesses.
4. *Hawk-Dove*. With $V = 4, C = 10$, find the ESS fraction of Hawks. *[0.4.]*
5. *Concept*. Give a symmetric game where (S, S) is Nash but S is not an ESS, and explain why.

Week 6 — Incomplete Information, Cooperation, and Coopetition

Three modules

Week 6 has three distinct parts. (1) **Bayesian games:** what happens when players have *private information* — and how auctions resolve it. (2) **Cooperative game theory:** when binding deals are possible, which coalitions form and how is the surplus split (the *core*)? (3) **Coopetition:** a managerial framework (the *Value Net* and *PAARTS*) for changing the game itself.

2.1 Bayesian Games (Incomplete Information, Static)

Complete vs. incomplete information

Complete information: every player knows all elements of the game (actions, payoffs, types). **Incomplete information:** at least one player has private knowledge — about her own *payoffs*, *characteristics* (“type”), or *available actions* — that others do not have. Such a player can try to *manipulate others’ beliefs* about her to shape the outcome.

2.1.1 Asymmetric information: signaling and screening

The two roles and their tools

Split players into **Better-Informed (BI)** and **Less-Informed (LI)**.

- A BI player can *conceal* or *mislead*, *reveal selectively but truthfully*, or **signal** private information *credibly* (a costly action the wrong type wouldn’t imitate).
- An LI player can **screen** (design choices that induce different types to self-select) or deliberately stay ignorant (*credible deniability*).

Why not simply ask?

Because **talk is cheap**. A BI player can claim anything; cheap talk carries no commitment, and she may even *signal-jam* (use a mixed strategy to confuse observers). Credible information transfer needs *costly* signals or cleverly designed *screens*, not mere assertions.

2.1.2 Modelling types: the Bayesian game and its equilibrium

Bayesian game and Bayesian Nash equilibrium

A **Bayesian game** adds, for each player i , a set of *types* Θ_i (her private information), a *prior* probability distribution over types, and payoffs that depend on the type profile. A **(pure) Bayesian Nash equilibrium** is a strategy for each *type* of each player such that every type is best-responding *in expectation* over the others' types, given the prior. In short: “each *type* maximises expected payoff given its beliefs.”

(Harsanyi's trick: model private information as a move by “Nature” that draws each player's type, after which the game is one of imperfect — but now well-defined — information.)

2.1.3 Worked example: Battle of the Sexes with incomplete information

Simple Intuition

Player 1 wants to coordinate with player 2, but is *unsure of player 2's mood*: with probability $\frac{1}{2}$ player 2 also wants to meet (BoS as usual), and with probability $\frac{1}{2}$ player 2 wants to *avoid* player 1. Player 2 knows her own type and player 1's preferences. This is a static game of incomplete information — and exactly the structure several Quiz-2 questions use.

The two type-contingent payoff matrices (player 1 chooses the row B/S ; entries are (u_1, u_2)):

	B	S
B	3, 1	0, 0
S	0, 0	1, 3

type “2 wants to meet” (prob $\frac{1}{2}$)

	B	S
B	3, 0	0, 3
S	0, 1	1, 0

type “2 wants to avoid” (prob $\frac{1}{2}$)

A *strategy* for player 2 is a pair (action if “meet”, action if “avoid”). Player 1, not knowing the type, evaluates *expected* payoffs (each type weight $\frac{1}{2}$).

Player 1's expected-payoff table and the Bayesian NE

Against player 2's type-contingent strategy $(\cdot, \cdot)$, player 1's expected payoff is the average of the two matrices:

	(B, B)	(B, S)	(S, B)	(S, S)
B	3	1.5	1.5	0
S	0	0.5	0.5	1

(e.g. row B , column (B, S) : $\frac{1}{2} \cdot 3 + \frac{1}{2} \cdot 0 = 1.5$.) Now solve for the Bayesian Nash equilibrium:

- “**Meet**” type of 2 wants to *match* player 1 (best response: B to B , S to S).
- “**Avoid**” type of 2 wants to *mismatch* (best response: S to B , B to S).
- **Player 1**, anticipating this: if she plays B , player 2's profile is (B, S) , giving her 1.5; if she plays S , the profile is (S, B) , giving her 0.5. She prefers B .

Bayesian NE: Player 1 plays B ; Player 2 plays $(B \text{ if meet, } S \text{ if avoid}) = (B, S)$.

★ Exam Tip

This is the template for the Quiz-2 “find a, b, c, d, p, q, r, t ” table: those letters are exactly the eight expected-payoff entries above ($a=3, b=1.5, c=1.5, d=0, p=0, q=0.5, r=0.5, t=1$), and the BNE is $(B, (B, S))$. Compute each cell as a $\frac{1}{2}/\frac{1}{2}$ average across the two type-matrices, then best-respond type by type.

2.1.4 First-price sealed-bid auction with uniform values

Simple Intuition

Each bidder writes a bid in a sealed envelope; the highest bidder wins and *pays her own bid*. You don't know your rival's value, only its distribution. Bidding your full value guarantees zero profit; shading your bid down trades a lower win-probability for a positive margin. Where is the sweet spot?

Continuous values and the uniform distribution

A value V on $[a, b]$ has cumulative distribution $F(x) = \Pr[V \leq x]$ and density $f(x) = F'(x)$. For the **uniform** distribution on $[0, 1]$: $F(x) = x$ and $f(x) = 1$, with $\mathbb{E}[V] = \frac{1}{2}$. “ $\Pr[\text{rival's value} < x] = x$ ” is the only fact we need below.

Equilibrium bidding: two bidders, values uniform on $[0, 1]$

Guess a symmetric linear strategy: each bidder with value v bids $b(v) = \alpha v$. A bidder of value v choosing bid b wins iff her bid beats the rival's, i.e. iff the rival's value w satisfies $\alpha w < b$, i.e. $w < b/\alpha$; since w is uniform, $\Pr[\text{win}] = b/\alpha$. Expected payoff:

$$\pi(b) = (v - b) \cdot \frac{b}{\alpha}.$$

Maximise over b : $\frac{d\pi}{db} = \frac{v - 2b}{\alpha} = 0 \Rightarrow b = \frac{v}{2}$. This is consistent with the guess if $\alpha = \frac{1}{2}$. Hence

$$b(v) = \frac{v}{2} \quad (\text{bid half your value}).$$

Bid-shading is the key lesson: with two bidders you optimally bid half your value; the more rivals, the less you shade.

General symmetric first-price equilibrium (uniform $[0, 1]$)

With n bidders whose values are i.i.d. uniform on $[0, 1]$, the symmetric equilibrium bid is

$$b(v) = \frac{n-1}{n} v.$$

($n = 2 \Rightarrow v/2$; as $n \rightarrow \infty$, $b(v) \rightarrow v$ — competition erodes the shading.)

★ Exam Tip

Two facts examiners love: (i) in a first-price auction you *shade below* your value, and the shading shrinks as bidders increase; (ii) by *revenue equivalence*, the seller's expected revenue here equals that of a second-price auction — where bidding your *true* value is a (weakly) dominant strategy. Keep “first-price = shade, second-price = truthful” straight.

2.1.5 The second-price (Vickrey) auction: why truth-telling dominates

Bidding your value is weakly dominant

In a **second-price** sealed-bid auction the highest bidder wins but pays the *second*-highest bid. Fix your value v and let h be the highest *rival* bid. Compare bidding your true value v against any other bid b :

- **Bidding above your value** ($b > v$): only changes the outcome when $v < h < b$ — you now win but pay $h > v$, a *loss*. Never helps.

- **Bidding below your value** ($b < v$): only changes the outcome when $b < h < v$ — you now *lose* an auction you'd have won at a profit. Never helps.

In every case bidding v does at least as well, so **truthful bidding is a (weakly) dominant strategy** — no shading, no guessing about rivals. This “strategy-proofness” is exactly why second-price/Vickrey rules underpin real ad auctions and the matching mechanisms of Week 7.

Revenue equivalence — the punchline

Despite their different rules, the first-price (shade to $\frac{n-1}{n}v$) and second-price (bid v , pay second-highest) auctions yield the seller the *same expected revenue* under the standard assumptions (risk-neutral bidders, i.i.d. private values). Intuitively the shading in the first-price auction exactly compensates for the fact that the winner pays her own (higher) bid. The *Revenue Equivalence Theorem* generalises this to a whole class of auctions.

2.1.6 Signaling vs. screening: a worked contrast

Education as a signal (Spence)

A worker is privately either high- or low-productivity. Talk is cheap, so a worker simply *claiming* “I’m high-productivity” is ignored. But if acquiring a credential is *cheaper for high types* than low types, a high type can **signal** by getting the credential at a cost the low type won’t imitate — a *separating* equilibrium where the credential credibly reveals type even if it teaches nothing. The mirror tool is **screening**: the *uninformed* employer offers a menu (e.g. pay-for-performance vs. flat wage) designed so different types *self-select*, revealing their type through choice. Signaling = informed party moves to reveal; screening = uninformed party designs choices to extract.

2.2 Cooperative Game Theory

A different question

Non-cooperative theory asks “what will each self-interested player *do*?” Cooperative theory assumes players can sign *binding* agreements and instead asks: *which coalitions form, and how should the jointly-created value be divided?* (Confusingly, there can be cooperation inside non-cooperative games and rivalry inside cooperative ones — the names refer to whether binding deals are enforceable.)

Cooperative game (N, v)

A **cooperative game** is a pair (N, v) : a set of players N and a **characteristic function** v assigning to every coalition (subset) $S \subseteq N$ the *value* $v(S)$ it can create on its own. The central object is how to divide $v(N)$, the value of the grand coalition.

2.2.1 A worked example: one seller, two buyers

Setup. $N = \{1, 2, 3\}$. Player 1 is a *seller* with one unit, cost Rs. 400. Player 2 (buyer) is willing to pay Rs. 900; Player 3 (buyer) is willing to pay Rs. 1100. Value is created only when the seller trades with a buyer (surplus = WTP – cost):

$$v(\{1, 2\}) = 900 - 400 = 500, \quad v(\{1, 3\}) = 1100 - 400 = 700,$$

$$v(\{1, 2, 3\}) = 700 \text{ (sell to the higher-value buyer 3)}, \quad v(\{1\}) = v(\{2\}) = v(\{3\}) = v(\{2, 3\}) = 0.$$

(Buyers alone, or two buyers without the seller, create nothing.)

Allocations and the criteria

An **allocation** $(x_1, \dots, x_n)$ gives player i the amount x_i . It is:

- **Individually rational** if $x_i \geq v(\{i\})$ for all i (no one accepts less than going alone);
- **Efficient** if $\sum_i x_i = v(N)$ (the whole pie is divided);
- Consistent with the **Marginal Contribution Principle** if $x_i \leq MC_i$, where $MC_i = v(N) - v(N \setminus \{i\})$ is how much i adds to the grand coalition.

Marginal contribution

$MC_i = v(N) - v(N \setminus \{i\})$. A player's bargaining power flows from (a) how badly the others *need* her and (b) her best outside option (**BATNA** — Best Alternative To a Negotiated Agreement).

Marginal contributions in the example

$$MC_1 = v(N) - v(\{2, 3\}) = 700 - 0 = 700, \quad MC_2 = v(N) - v(\{1, 3\}) = 700 - 700 = 0, \quad MC_3 = v(N) - v(\{1, 2\}) = 700 - 500 = 200$$

Buyer 2 adds *nothing* (the seller would rather deal with buyer 3 anyway), so competition between buyers hands power to the seller.

2.2.2 The Core

The Core

An efficient allocation $(x_1, \dots, x_n)$ is in the **core** if *no coalition can do better on its own*: for every $S \subseteq N$,

$$\sum_{i \in S} x_i \geq v(S).$$

Equivalently, no subset of players has the incentive and ability to break away and split a larger value among themselves. The core can be empty, a single point, or (often) a range.

Computing the core of the seller/buyers game

Efficiency: $x_1 + x_2 + x_3 = 700$. Coalition constraints:

$$x_1 + x_3 \geq v(\{1, 3\}) = 700, \quad x_1 + x_2 \geq v(\{1, 2\}) = 500, \quad x_i \geq 0.$$

From $x_1 + x_3 \geq 700$ and the total = 700, we get $x_2 \leq 0$, hence $\mathbf{x_2 = 0}$ (buyer 2 gets nothing). Then $x_1 + x_3 = 700$, and $x_1 + x_2 \geq 500 \Rightarrow x_1 \geq 500$; also $x_3 = 700 - x_1 \geq 0 \Rightarrow x_1 \leq 700$. Therefore

$$\text{Core} = \{(x_1, 0, 700 - x_1) : 500 \leq x_1 \leq 700\}.$$

The seller captures between 500 and 700; buyer 3 gets between 0 and 200 (his marginal contribution); buyer 2 gets 0. **Competition between the buyers transfers nearly all surplus to the seller** — the core makes “added value = bargaining power” precise.

Equilibrium Insight

Notice the core respects the marginal-contribution ceilings: $x_2 \leq MC_2 = 0$ and $x_3 \leq MC_3 = 200$. “You can’t capture more than you add” is the cooperative-game version of competitive pressure. (The *Shapley value* — a unique “fair” point rather than a range —

is the natural next concept; it is treated in Book 3.)

Preview: from a *range* (core) to a *point* (Shapley value)

The core often gives a whole *interval* of stable splits, which is unsatisfying if you need one fair number. The **Shapley value** resolves this by paying each player her *average marginal contribution* over all orders in which the coalition could form:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (|N| - |S| - 1)!}{|N|!} [v(S \cup \{i\}) - v(S)].$$

It is the unique allocation satisfying efficiency, symmetry, the dummy property, and additivity. (Full treatment in Book 3 — but recognise that the core's marginal-contribution ceilings are exactly the raw material the Shapley value averages.)

	Non-cooperative	Cooperative
Binding deals?	No (self-enforcing only)	Yes (enforceable contracts)
Primitive	Strategies & payoffs	Coalition values $v(S)$
Question	What will each player do?	How is $v(N)$ divided?
Solution	Nash / SPE / BNE	Core, Shapley value
Power from	Best responses	Marginal contribution / BATNA

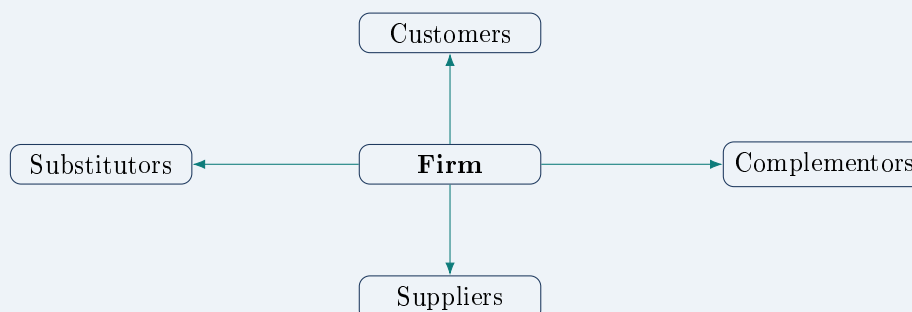
2.3 Coopetition and the Right Game

Simple Intuition

Brandenburger & Nalebuff's **coopetition** = cooperation + competition. Business is a game with no fixed rulebook: you *cooperate to grow the pie* and *compete to grab a larger slice*. The strategic move is often not to play the given game better, but to **change the game**.

The Value Net

Map every player around a firm:



Vertical axis: customers (above) and suppliers (below) — the value chain. **Horizontal** axis: *substitutors* (competitors) and *complementors*. The same firm can be both a foe *and* a friend.

Competitor vs. complementor

- A firm is your **complementor** (in customers) if customers value your product *more* when they also have the other's: $WTP(A \& B) > WTP(A) + WTP(B)$. (In suppliers: it's *more* attractive to supply you when also supplying the other: $OC(A \& B) < OC(A) + OC(B)$.)
- A firm is your **competitor/substitutor** if the inequalities reverse (having the other makes you *less* valuable, or makes supplying you *less* attractive).

★ PAARTS: the six levers to change the game

To change a game you must change at least one element — **PAARTS**:

- **Players** — add/remove players (e.g. Coke encouraging Holland Sweetener to enter aspartame, weakening Monsanto's hold).
- **Actions** — enlarge or restrict the action set.
- **Added value** — raise your own or lower a rival's (Nintendo *withheld* consoles to keep its added value high and capture more profit).
- **Rules** — simultaneous vs. sequential, contracts, norms ("look forward and reason back" to anticipate reactions).
- **Tactics** — manage perceptions: signal, or maintain/remove the "fog."
- **Scope** — expand or shrink the boundaries of the game (link or separate markets).

Equilibrium Insight

The deep idea: "*You can't take away more than you add.*" Your sustainable share is bounded by your *added value*, so smart strategy actively *shapes* added value, players, and rules rather than passively optimising within a fixed game. Win-lose moves (cut price for share) often backfire when rivals match (lose-lose); win-win moves grow the pie for everyone.

Added value, made concrete: the card game

Brandenburger & Nalebuff's classic illustration: a referee has 26 black cards; 26 students each hold one red card; each black-red pair redeemed earns \$100. Total pie = $26 \times \$100 = \2600 . **Added value** of the referee = whole pie (without him no pair forms) = \$2600; added value of *each* student = \$100 (removing one student loses exactly one pair). Sum of all added values ($2600 + 26 \times 100 = 5200$) *exceeds* the pie — so added values are *ceilings*, not entitlements, and bargaining decides the split within them. Now suppose the referee *burns 3 black cards*: he creates scarcity, raising each remaining student's competition. With 23 black and 26 red, 3 students are now "excess" — their added value drops toward \$0, shifting bargaining power to the referee. This is the **Players** and **Added value** levers of PAARTS in action (echoing the seller/buyers core result: scarcity on one side transfers surplus).

End of Week 6**Quick Revision — Week 6**

Bayesian games: private types + prior; each *type* best-responds in expectation (Bayesian NE). Cheap talk is non-credible; use signals/screens. **First-price auction (uniform $[0, 1]$)**: bid $b(v) = \frac{n-1}{n}v$ (shade; $v/2$ for two bidders). **Cooperative game (N, v)** : characteristic function. $MC_i = v(N) - v(N \setminus i)$. **Core**: efficient allocation with $\sum_{i \in S} x_i \geq v(S)$ for all S . **Coopetition**: Value Net (customers/suppliers/substitutors/complementors); change the game via **PAARTS**.

Formula Sheet — Week 6

$$\Pr[V < x] = F(x) \text{ (} = x \text{ for uniform)}; \quad b(v) = \frac{n-1}{n}v; \quad \text{MC}_i = v(N) - v(N \setminus \{i\}).$$

Core: $\sum_{i \in N} x_i = v(N)$ and $\sum_{i \in S} x_i \geq v(S) \forall S$. **Complementor (customers):** $\text{WTP}(A \& B) > \text{WTP}(A) + \text{WTP}(B)$.

★ Algorithm sheet

First-price bid: assume $b = \alpha v \rightarrow$ write $\Pr[\text{win}] \rightarrow$ maximise $(v - b) \Pr[\text{win}] \rightarrow$ solve.

Core: write efficiency + all coalition inequalities $\rightarrow$ deduce tight constraints $\rightarrow$ express the feasible range. **MC ceilings:** compute $v(N) - v(N \setminus i)$ to bound each player's share.

Practice questions.

1. *Auction.* With three bidders (uniform $[0, 1]$), what is the equilibrium bid function? $b(v) = \frac{2}{3}v$.
2. *Core.* If buyer 3's WTP rises to 1300, recompute $v(\{1, 3\})$, $v(N)$ and the core.
3. *MC.* Show $x_2 = 0$ in the example follows directly from $\text{MC}_2 = 0$.
4. *Coopetition.* Classify: is a smartphone maker a complementor or competitor to an app developer? Justify with the WTP inequality.
5. *Concept.* Why is "just ask the informed player" usually ineffective? Name the remedy.

Week 7 — Matching and the Gale–Shapley Algorithm

The problem

How do you pair up two groups — doctors and hospitals, students and schools, organ donors and recipients, men and women — so that the result is *stable*: nobody can find a partner they'd both rather be with? Gale & Shapley (1962) defined stability, proved a stable matching always exists (for the two-sided problem), and gave an algorithm to find it. The work won Shapley and Roth the 2012 Nobel Prize and underlies real systems like the medical residency match.

3.1 The Matching Problem

Setup (marriage model)

Equal numbers of men and women. Each person submits a strict *ranking* of all members of the opposite group (no indifference, for now). A **matching** pairs each man with exactly one woman. We want a matching that the matchmaker's recommendation will actually *survive* — one no pair wants to overturn.

Blocking pair and stability

A pair (m, w) *not* matched to each other is a **blocking pair** if m prefers w to his assigned partner *and* w prefers m to hers — they would elope. A matching is **stable** if it has *no blocking pair*. (Equivalently: no unmatched man–woman couple prefers each other to their current mates.)

Why a “feasible” matching can fail: a blocking pair

Suppose a proposal pairs Cher with her *last* choice while Al ranks Cher *first*. If Cher also prefers Al to her assigned mate, then (Al, Cher) is a blocking pair: they reject the matchmaker and pair up. Any matching containing such a couple is **unstable** — it will be torn apart in practice. Pairing “everyone with their first choice” usually fails because two people often share a top choice, leaving conflicts that generate blocking pairs.

Stable $\neq$ everyone happy

A stable matching need not make anyone's *first* choice come true; it only guarantees that no *pair* can *jointly* improve. If one side gets its first choice, the other side — however un-

happy — cannot undermine the match (the desired partners are already happily matched and will say no). Stability serves the *matchmaker*, not necessarily every individual.

3.2 Existence Is Not Automatic: the Roommate Problem

A problem with *no* stable matching

Split four boys $\{a, b, c, d\}$ into two rooms. Preferences:

$$a : b \succ c \succ d, \quad b : c \succ a \succ d, \quad c : a \succ b \succ d, \quad d : \text{(last for everyone)}.$$

Check all three pairings:

- $(a, b), (c, d)$: b prefers c , and c prefers b to $d \Rightarrow \{b, c\}$ block.
- $(a, c), (b, d)$: a prefers b , and b prefers a to $d \Rightarrow \{a, b\}$ block.
- $(a, d), (b, c)$: a prefers c , and c prefers a to $b \Rightarrow \{a, c\}$ block.

Every matching has a blocking pair: **no stable matching exists**. The two-sided structure of the marriage problem (men vs. women) is exactly what guarantees existence; the one-sided roommate problem can fail.

3.3 The Gale–Shapley Algorithm (Deferred Acceptance)

The men-proposing algorithm

Repeat in stages until no man is rejected:

1. **Propose**. Every man not currently held proposes to the highest woman on his list who has not yet rejected him.
2. **Tentatively hold / reject**. Every woman who has ≥ 2 current proposers (including anyone on her “waiting list”) keeps her *favourite* on a waiting list and *rejects* the rest.
3. **Iterate**. Rejected men propose down their lists next stage.

When no man is rejected, every man on a waiting list marries that woman. (Women’s acceptances are *tentative* — “deferred” — until the end, which is why it’s called *deferred acceptance*.)

A small run with a rejection

Men $\{A, B\}$, women $\{a, b\}$. Preferences $A : a \succ b, B : a \succ b; a : B \succ A, b : A \succ B$.

- **Stage 1**. $A \rightarrow a, B \rightarrow a$. Woman a holds her favourite B , *rejects* A .
- **Stage 2**. Rejected A proposes to b ; b holds A .

No one is rejected $\Rightarrow$ terminate. Matching $(A, b), (B, a)$. Check: no blocking pair, so it’s stable.

Guaranteed properties

GS-P1 (termination)

The algorithm stops after *finitely many* stages (each rejected man moves strictly down his finite list).

GS-P2 (stability)

The matching it produces is always *stable*. Hence a stable matching *always exists* for the marriage problem.

Why the output is stable (proof sketch)

Suppose, for contradiction, (m, w) is a blocking pair: m prefers w to his assigned w' , and w prefers m to her assigned m' . Since m prefers w to w' , he must have proposed to w *before* settling for w' (men go down their lists in order). So w either rejected m then, or accepted him then dropped him later — in both cases only because she held *someone she prefers to m* . Women only ever trade up, so her *final* partner m' is at least as good *to her* as m — contradicting “ w prefers m to m' .” No blocking pair can exist.

Complexity and termination count

Each man proposes to each woman at most once, so there are at most n^2 proposals total: the algorithm runs in $O(n^2)$ time and always halts. A useful corollary: at most $n^2 - n + 1$ proposal-rounds occur in the worst case.

Strategy-proofness (a deep, exam-worthy fact)

In the *proposing*-side-optimal mechanism, **no proposer can do better by lying** about her preferences — truthful ranking is a (weakly) dominant strategy for the proposing side (Roth). The *receiving* side, however, *can* sometimes gain by misrepresenting. This asymmetry, plus proposer-optimality (GS-P3), is why designing “who proposes” is the central policy choice in real clearinghouses.

3.4 Multiplicity and Optimality

A preference structure with *three* stable matchings

Men $\{A, B, C\}$, women $\{a, b, c\}$:

$A : a \succ c \succ b, \quad B : b \succ a \succ c, \quad C : c \succ b \succ a; \quad a : C \succ B \succ A, \quad b : A \succ C \succ B, \quad c : B \succ A \succ C.$

There are *three* stable matchings:

1. Men get their first choice: $(A, a), (B, b), (C, c)$.
2. Women get their first choice: $(A, b), (B, c), (C, a)$.
3. Everyone gets their *second* choice: $(A, c), (B, a), (C, b)$.

Men-proposing GS (stage 1: $A \rightarrow a, B \rightarrow b, C \rightarrow c$ — all accepted) yields matching 1, the *men-optimal* one. **Women-proposing GS** (stage 1: $a \rightarrow C, b \rightarrow A, c \rightarrow B$) yields matching 2, the *women-optimal* one. Matching 3 is stable but produced by neither one-sided run.

Optimality for a side

A stable matching is **optimal for the men** if every man is at least as well off under it as under *any* other stable matching (and similarly for women). We compare only *among stable matchings*.

The distributional punchline**GS-P3A**

The matching from the *men-proposing* algorithm is **optimal for every man** (and, simultaneously, *worst* for every woman among stable matchings).

GS-P3B

The *women-proposing* algorithm is optimal for every woman.

The side that proposes gets its best stable outcome. Who holds the proposing

role has real distributional consequences — a key design lever in real matching markets.

★ Exam Tip

Three questions to keep straight: (1) *Does a stable matching exist?* Always, for the two-sided problem (GS-P2); not always for roommates. (2) *Is it unique?* No — there can be several. (3) *Is one everyone's favourite?* No, but there *is* a men-best and a women-best one. “Proposer-optimal” is the most-tested fact.

3.5 Generalizations

Relaxing the assumptions

The algorithm extends naturally:

- **Unequal numbers / incomplete lists:** some people stay unmatched; a proposer skips anyone not on his list, and a receiver rejects unacceptable proposers. Run GS as usual (e.g. it may return only $(A, d), (B, a)$ with others unmatched).
- **Ties (indifference):** break ties arbitrarily to get strict lists, then run GS. Stability is defined so that a person won't leave a partner for one they're *indifferent* to (only a *strict* improvement blocks).
- **Many-to-one (college admissions / assignment):** each college has a *quota* q ; it holds its top q applicants and rejects the rest. This is the practically important version — assigning students to schools, doctors to hospitals.

Why the procedural setup matters in practice

Real admissions face exactly the failure modes GS fixes: students manipulate by feigning interest, colleges over-admit fearing no-shows, and the process becomes a guessing game with enrolments far from optimal. A deferred-acceptance clearinghouse produces a *stable* assignment and (in the applicant-proposing form) makes truthful ranking a safe strategy for applicants.

★ Where this is actually used (Roth's Nobel work)

- **NRMP** — the U.S. National Resident Matching Program assigns graduating doctors to hospitals via (doctor-proposing) deferred acceptance.
 - **School choice** — Boston and New York City public-school assignment were redesigned around stable/strategy-proof deferred-acceptance mechanisms.
 - **Kidney exchange** — chains and cycles of donor–patient pairs are matched with the same stability logic (no pair wants to defect to a better swap).
- Shapley (theory) and Roth (practical market design) shared the 2012 Nobel Prize for exactly this.

End of Week 7

Quick Revision — Week 7

Stable matching: no *blocking pair* (no two people who'd both rather be together).
Roommate problem: a stable matching may *not* exist (one-sided). **Gale–Shapley (deferred acceptance):** proposers propose down their lists; receivers tentatively hold the best and reject the rest; repeat. **GS-P1** terminates; **GS-P2** output is stable (so

existence is guaranteed for two-sided); **GS-P3** the *proposing* side gets its optimal stable matching. Stable $\neq$ everyone happy.

★ Algorithm sheet — running Gale–Shapley

(1) Pick the proposing side. (2) Each free proposer proposes to its top *not-yet-rejected* choice. (3) Each receiver keeps its favourite among current + new proposers, rejects the rest. (4) Repeat until no rejections. (5) Read off the matching; verify stability by scanning for any blocking pair. **Tip:** men-proposing $\Rightarrow$ men-optimal/women-pessimal; swap roles for the other extreme.

Property	Marriage (two-sided)	Roommate (one-sided)
Stable matching exists?	Always (GS-P2)	Not always
Found by	Gale–Shapley	—
Unique?	Often no	—
Proposer effect	Proposer-optimal	—

Practice questions.

1. *Run GS.* With $A : a \succ b$, $B : a \succ b$, $a : A \succ B$, $b : A \succ B$, run men-proposing and women-proposing. Are they the same?
2. *Blocking pair.* Given a proposed matching, describe the exact check for whether (m, w) blocks it.
3. *Optimality.* In the three-stable-matching example, confirm matching 1 is men-optimal by comparing each man's partner across the three.
4. *Roommate.* Construct your own 4-person roommate instance with no stable matching.
5. *Concept.* Explain why a stable matching need not maximise the number of people getting their first choice.

Week 8 — Information Cascades and Voting

Two faces of collective behaviour

When people decide *in sequence* and watch each other, they may rationally *ignore their own information and copy the crowd* — an **information cascade**. When a group must instead *aggregate* conflicting preferences into one decision, every voting rule turns out to have pathologies — culminating in **Arrow's Impossibility Theorem**. Week 8 covers both.

4.1 Information Cascades (Herding)

The restaurant

You researched restaurant A, but arrive to find A empty and B next door packed. If other diners have similar tastes and their own information, the *crowd* at B may be stronger evidence than your private research — so you rationally join B. When people abandon private signals in favour of inferences from earlier actions, **herding / an information cascade** has occurred.

Ingredients of a cascade

A cascade can form when: (A) there is a decision to make; (B) people decide *sequentially*, each *observing earlier choices*; (C) each has a *private signal*; (D) others' private signals are *not* directly observable but can be *inferred* from their actions. Fashions, fads, bestseller lists, technology adoption and bandwagon voting are all examples.

4.1.1 The urn experiment

Setup. An urn is equally likely to be *majority-blue* (2 blue, 1 red) or *majority-red* (2 red, 1 blue). One by one each student privately draws a marble (with replacement), then *publicly guesses* the urn's majority. Correct guesses are rewarded. Each reasons optimally from everything heard so far.

How the cascade starts

- **Student 1** should guess the colour drawn (one signal). His guess *reveals* his draw.
- **Student 2**, if she draws the same colour, guesses it; if opposite, she's indifferent (one of each) and breaks the tie toward her own draw — so her guess also reveals her draw.

- **Student 3:** if the first two guesses *differ*, he follows his own draw. But if the first two are the *same* (say both blue) and he draws red, he reasons he has effectively seen blue, blue, red $\Rightarrow$ still majority-blue, so he **guesses blue regardless of his own draw**. His guess now conveys *no* information.
- **Student 4 onward:** knowing student 3's guess was uninformative, she faces the same two informative “blue” signals plus her own — and also guesses blue no matter what. **The cascade is locked in.**

4.1.2 Bayes' rule makes it precise

Bayes' rule

$$\Pr[A \mid B] = \frac{\Pr[A \cap B]}{\Pr[B]} = \frac{\Pr[B \mid A] \Pr[A]}{\Pr[B]}.$$

A guesser should choose “majority-blue” iff $\Pr[\text{majority-blue} \mid \text{all she saw/heard}] > \frac{1}{2}$.

The numbers behind the cascade

Priors $\Pr[\text{maj-blue}] = \Pr[\text{maj-red}] = \frac{1}{2}$; signal accuracy $\Pr[\text{blue} \mid \text{maj-blue}] = \Pr[\text{red} \mid \text{maj-red}] = \frac{2}{3}$.

After one blue draw: $\Pr[\text{blue}] = \frac{1}{2} \cdot \frac{2}{3} + \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{2}$, so

$$\Pr[\text{maj-blue} \mid \text{blue}] = \frac{\frac{1}{2} \cdot \frac{2}{3}}{\frac{1}{2}} = \frac{2}{3} > \frac{1}{2} \Rightarrow \text{guess blue.}$$

After blue, blue, red (third student draws red): draws are independent, so

$$\Pr[bbr \mid \text{maj-blue}] = \frac{2}{3} \cdot \frac{2}{3} \cdot \frac{1}{3} = \frac{4}{27}, \quad \Pr[bbr \mid \text{maj-red}] = \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{2}{3} = \frac{2}{27},$$

$$\Pr[bbr] = \frac{1}{2} \left(\frac{4}{27} + \frac{2}{27} \right) = \frac{1}{9}, \quad \Pr[\text{maj-blue} \mid bbr] = \frac{\frac{1}{2} \cdot \frac{4}{27}}{\frac{1}{9}} = \frac{2}{3} > \frac{1}{2}.$$

The student **rationally guesses blue despite drawing red** — the two earlier blue signals outweigh her own. The cascade is mathematically justified.

Properties of cascades

Cascades can be wrong: the first two (possibly mistaken) signals can lock everyone into the incorrect urn. **Cascades are fragile:** a small amount of new *public* information can shatter them. They explain why fads, bubbles, and bestseller dynamics can be both powerful and suddenly reversible — and why early movers have outsized influence.

★ Variant: the “3-and-1” urn (used in Quiz 2)

Suppose instead each urn holds *four* balls: majority-blue = {3 blue, 1 red}, majority-red = {3 red, 1 blue}, equal priors. Then $\Pr[\text{blue} \mid \text{maj-blue}] = \frac{3}{4}$ and $\Pr[\text{blue} \mid \text{maj-red}] = \frac{1}{4}$ (the signal is *stronger* than in the 2-1 urn). The method is identical:

- One draw from the red-majority urn is blue with probability $\frac{1}{4}$.
- After a single red draw: $\Pr[\text{maj-red} \mid \text{red}] = \frac{\frac{1}{2} \cdot \frac{3}{4}}{\frac{1}{2}} = \frac{3}{4}$ (so a lone red draw already points to red-majority; guessing blue would be wrong).
- Sequence (red, red, blue): $\Pr[rrb \mid \text{maj-blue}] = \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{64}$, $\Pr[rrb \mid \text{maj-red}] =$

$$\frac{3}{4} \cdot \frac{3}{4} \cdot \frac{1}{4} = \frac{9}{64}, \text{ total } \Pr[rrb] = \frac{1}{2}(\frac{3}{64} + \frac{9}{64}) = \frac{3}{32}, \text{ and } \Pr[\text{maj-red} \mid rrb] = \frac{\frac{1}{2} \cdot \frac{9}{64}}{\frac{3}{32}} = \frac{3}{4}.$$

So after two reds the third student rationally announces red *even after drawing blue* — the cascade forms exactly as before, just with $\frac{3}{4}/\frac{1}{4}$ in place of $\frac{2}{3}/\frac{1}{3}$.

4.2 Voting and Social Choice

Individual preferences

A group must turn individual rankings of ≥ 2 alternatives into a single *group ranking*. Each individual's preference $\succ_i$ must be:

- **[IP1] Complete:** for any distinct X, Y , exactly one of $X \succ_i Y$ or $Y \succ_i X$ (no ties, no blanks).
- **[IP2] Transitive:** $X \succ_i Y$ and $Y \succ_i Z \Rightarrow X \succ_i Z$.

(Intransitive tastes — chocolate $\succ$ vanilla $\succ$ kesar-pista $\succ$ chocolate — leave you unable to choose, which is why we forbid them at the individual level.)

Voting system

A **voting system** is any function taking a profile of complete, transitive individual rankings and returning a group ranking. With *two* alternatives, the standard choice is **majority rule** (assume an odd number of voters to avoid ties).

4.2.1 The Condorcet Paradox

Majority rule can produce *intransitive* group preferences

Three voters, three alternatives:

$$1: A \succ B \succ C, \quad 2: B \succ C \succ A, \quad 3: C \succ A \succ B.$$

Pairwise majorities: $A \succ B$ (voters 1,3), $B \succ C$ (voters 1,2), $C \succ A$ (voters 2,3). So the group has $A \succ B \succ C \succ A$ — a **cycle**, violating transitivity even though every *individual* is transitive. This is the **Condorcet Paradox**.

Agenda-setting power

With a cyclic group preference, whoever *sets the agenda* of pairwise votes controls the winner. In the example: to make C win, first vote A vs B (then winner A loses to C); to make A win, first vote B vs C ; to make B win, first vote A vs C . The “group favourite” is decided by the sequence, not the preferences — and re-introducing eliminated options never terminates.

4.2.2 Positional voting: Borda count

Borda count

With k alternatives, each voter's ranking awards $k - 1$ points to her top choice, $k - 2$ to the next, $\dots$, 0 to her last. Sum across voters; rank by total points. Borda *always* yields a complete, transitive group ranking (it sorts by a single number).

Borda example

2 voters, 4 alternatives, points 3, 2, 1, 0:

Voter 1: $A \succ B \succ C \succ D$, Voter 2: $B \succ C \succ A \succ D$.

$A = 3 + 1 = 4$, $B = 2 + 3 = 5$, $C = 1 + 2 = 3$, $D = 0$. Group ranking: $B \succ A \succ C \succ D$.

Borda violates “independence of irrelevant alternatives”

3 experts rank $\text{Pele} \succ \text{Maradona} \succ \text{Messi}$; 2 rank $\text{Maradona} \succ \text{Messi} \succ \text{Pele}$ (points 2, 1, 0):

$\text{Pele} = 3(2) + 2(0) = 6$, $\text{Maradona} = 3(1) + 2(2) = 7$, $\text{Messi} = 3(0) + 2(1) = 2$.

Borda crowns **Maradona** — yet *head-to-head* Pele beats Maradona 3 to 2! Adding the “irrelevant” option Messi flipped the Pele–Maradona result. **Strategic misreporting**: if the 2 experts truly preferred $\text{Maradona} \succ \text{Pele} \succ \text{Messi}$, honest reporting would give Pele 8, Maradona 7 (Pele wins); by *lying* (sinking Pele below Messi) they make Maradona win. Borda rewards strategic dishonesty.

4.2.3 Other systems — and how they disagree**One profile, three different winners**

Budget to one of Health/Security/Education; 12 voters: 3 rank $H \succ S \succ E$; 4 rank $E \succ H \succ S$; 5 rank $S \succ E \succ H$.

- **Plurality** (most first-place): $S = 5$, $E = 4$, $H = 3 \Rightarrow$ **Security wins** — though a majority (7 of 12) prefer Health to Security.
- **Instant runoff**: no majority; eliminate fewest-first-place (Health, 3); Health’s voters ($H \succ S \succ E$) shift to Security $\Rightarrow$ Security = 8, **Security wins**.
- **Pairwise (Condorcet) check**: H beats S (7–5), S beats E (8–4), E beats H (9–3) — a *cycle*, so no Condorcet winner; methods that look for one (Copeland) cannot single out a clear champion.

Different reasonable rules pick different winners from the *same* preferences — the pathology Arrow’s theorem will explain. (*Copeland’s appeal*: if it yields a single winner, that winner beats every other head-to-head — a Condorcet winner — but here none exists.)

4.2.4 Arrow’s Impossibility Theorem**The two “no-pathology” axioms**

- **[VP1] Unanimity / Pareto**: if *every* individual prefers X to Y , the group ranking has $X \succ Y$.
- **[VP2] Independence of Irrelevant Alternatives (IIA)**: the group’s ordering of X vs. Y depends *only* on how individuals rank X vs. Y — shifting some third option Z must not flip X vs. Y .

IIA failures caused every pathology above (the Messi effect; strategic agenda-setting).

Arrow’s Impossibility Theorem

If there are at least three alternatives, the *only* voting system satisfying both **Unanimity** and **IIA** is a dictatorship — the group ranking always equals one fixed individual’s ranking. Equivalently: **no voting system satisfies Unanimity, IIA, and Non-dictatorship simultaneously** (for ≥ 3 alternatives). Any reasonable rule must

therefore sacrifice at least one desirable property — voting involves *unavoidable* trade-offs. (With only *two* alternatives, majority rule satisfies everything — the impossibility is special to ≥ 3 options.)

★ Exam Tip

Why does dictatorship “pass”? A dictator’s ranking trivially respects unanimity (if all prefer X to Y , so does the dictator) and IIA (the group’s X -vs- Y depends only on the dictator’s X -vs- Y , never on Z). Arrow’s theorem says this degenerate rule is the *only* escape — which is precisely why it’s an *impossibility* result for *democratic* aggregation.

Proof idea: the pivotal (decisive) voter

The standard proof builds a dictator in three moves. (1) *Extremal lemma*: if every voter puts some option B at the top *or* bottom of their list, society must too (by unanimity and IIA). (2) *Pivotal voter*: march voters one by one from “ B bottom” to “ B top”; at some voter n^* society’s ranking of B flips — this voter is *pivotal* for B . (3) *Spreading dictatorship*: using IIA one shows this same n^* dictates *every* pairwise comparison not involving B , then B too. Hence n^* is a dictator. The engine throughout is IIA, which forbids society from using any information beyond the pairwise rankings.

Escape routes (how real systems dodge Arrow)

Arrow assumes *unrestricted* preferences over ≥ 3 options with *ordinal* information. Relax an assumption and possibility returns:

- **Two alternatives only**: *May’s Theorem* — majority rule is the *unique* rule that is anonymous, neutral, and positively responsive. (Arrow needs ≥ 3 .)
- **Single-peaked preferences** (options on a line, each voter has one ideal point): the *Median Voter Theorem* — the *median voter’s* favourite is a Condorcet winner and majority rule is transitive. No cycles. (This is why Hotelling’s two parties converge to the median voter.)
- **Cardinal information** (intensities, not just rankings): rules like range/score voting sidestep IIA — at the cost of inviting exaggeration.

Gibbard–Satterthwaite: the strategic cousin of Arrow

A companion impossibility: for ≥ 3 alternatives, *every* non-dictatorial voting rule that can elect any candidate is **manipulable** — some voter can sometimes get a better outcome by *misreporting* preferences (exactly the Borda “strategic misreporting” you saw). Truthful voting cannot be made dominant for everyone. Arrow = no perfect *aggregation*; Gibbard–Satterthwaite = no perfect *strategy-proofness*.

System	How it decides	Main weakness
Plurality	Most first-place votes	Ignores all but first choice; “spoilers”
Borda	Weighted points $k-1, \dots, 0$	Violates IIA; manipulable
Condorcet	Beats all others pairwise	May not exist (cycles)
Instant runoff	Eliminate lowest, transfer	Non-monotonic; can skip Condorcet winner
Dictatorship	One person decides	Satisfies Arrow — but undemocratic

4.2.5 Voting to aggregate information

Simple Intuition

Sometimes voters share a *goal* (a jury seeking the truth) and differ only because they have different *information*. Here voting is less about reconciling tastes than *aggregating signals* — linking back to cascades: if jurors simply copy early votes, private information is lost, exactly as in a herding cascade. Good mechanisms must encourage voters to act on their private signals.

End of Week 8

Quick Revision — Week 8

Information cascade: sequential decisions + observed actions + private signals $\Rightarrow$ people rationally copy the crowd and ignore own info. **Bayes:** $\Pr[A | B] = \Pr[B | A] \Pr[A] / \Pr[B]$; in the urn, after two like guesses you follow the crowd ($\Pr = \frac{2}{3}$). Cascades can be wrong and are fragile. **Voting:** individual prefs complete & transitive; *majority rule* can be intransitive (**Condorcet paradox**). **Borda** violates IIA and invites strategic misreporting. **Arrow:** with ≥ 3 alternatives, Unanimity + IIA $\Rightarrow$ dictatorship (no system has all three of unanimity, IIA, non-dictatorship).

Formula Sheet — Week 8

$$\Pr[A | B] = \frac{\Pr[B | A] \Pr[A]}{\Pr[B]}; \quad \text{Borda points} = (k-1, k-2, \dots, 1, 0).$$

Condorcet cycle: $A \succ B$, $B \succ C$, $C \succ A$ from transitive individuals. **Arrow** (≥ 3 alts): Unanimity + IIA + Non-dictatorship is impossible.

★ Algorithm sheet

Cascade: write priors and signal accuracies $\rightarrow$ apply Bayes after each observation $\rightarrow$ a guess is a cascade once it no longer depends on the private draw. **Condorcet:** do all pairwise majority votes $\rightarrow$ look for a cycle / Condorcet winner. **Borda:** assign $k-1, \dots, 0$, sum, rank. **Spot the pathology:** check IIA (does an irrelevant option flip a pairwise result?).

Practice questions.

1. *Bayes.* After guesses blue, blue and a red draw, verify $\Pr[\text{maj-blue}] = \frac{2}{3}$.
2. *Condorcet.* Build a 3-voter profile with a cycle and show how an agenda-setter picks any winner.
3. *Borda.* Recompute the Pele/Maradona/Messi totals and confirm the IIA violation.
4. *Systems.* For the Health/Security/Education profile, find each system's winner.
5. *Arrow.* State Arrow's theorem and explain why dictatorship is the lone system passing unanimity + IIA.

Part II

Complete PYQ Solutions

How Part II Works

Across the four Quiz-2 papers (2024 T2/T3, 2025 T2/T3) the same *question families* recur with changed numbers. Each is solved here in the six-step style: **interpretation** → **topic** → **thought process** → **full working** → **traps** → **verified answer**. Answer keys were read directly from the papers and independently re-derived.

★ The Quiz-2 families

(1) **Information cascade** on a “3-and-1” urn (Bayes); (2) **Voting** (plurality, Borda); (3) **Gale–Shapley** matching; (4) **Repeated games** (finite PD; infinitely-repeated cooperation with discounting); (5) **Bayesian Battle of Sexes**; (6) **ESS** in 2×2 games. Master these six and you have seen the whole exam.

Family 1 — Information Cascade (the “3-and-1” Urn)

In every paper

A modified urn: majority-blue = {3 blue, 1 red} or majority-red = {3 red, 1 blue}, equal priors. A chain of Bayes computations follows.

5.1 Topic & key numbers

Week 8 (cascades, Bayes’ rule). Signal strengths: $\Pr[\text{matching colour} \mid \text{its majority}] = \frac{3}{4}$, $\Pr[\text{off colour}] = \frac{1}{4}$. Priors $\frac{1}{2}$ each.

5.2 Full working (all sub-question types)

1. “Pick from the red-majority urn — $\Pr[\text{blue}]?$ ” The red-majority urn has 1 blue of 4 $\Rightarrow \boxed{\frac{1}{4}}$. (Symmetrically, $\Pr[\text{red}]$ from blue-majority = $\frac{1}{4}$.)
2. “ $\Pr[\text{red-majority} \mid \text{first draw red}]?$ ” $\Pr[\text{red}] = \frac{1}{2} \cdot \frac{3}{4} + \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{2}$, so

$$\Pr[\text{maj-red} \mid \text{red}] = \frac{\frac{1}{2} \cdot \frac{3}{4}}{\frac{1}{2}} = \boxed{\frac{3}{4}}.$$

3. “Should student 1 guess *blue*-majority after drawing red?” No — the posterior favours red-majority ($\frac{3}{4} > \frac{1}{2}$), so the rational guess is red-majority. $\boxed{\text{No.}}$
4. Sequence (red, red, blue): $\Pr[\text{seq} \mid \text{blue-maj}] = \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{3}{4} = \boxed{\frac{3}{64}}$.
5. $\Pr[\text{seq} \mid \text{red-maj}] = \frac{3}{4} \cdot \frac{3}{4} \cdot \frac{1}{4} = \boxed{\frac{9}{64}}$.
6. Total $\Pr[\text{seq}] = \frac{1}{2} \left(\frac{3}{64} + \frac{9}{64} \right) = \frac{1}{2} \cdot \frac{12}{64} = \boxed{\frac{3}{32}}$.
7. $\Pr[\text{red-majority} \mid \text{red, red, blue}] = \frac{\frac{1}{2} \cdot \frac{9}{64}}{\frac{3}{32}} = \frac{9/128}{3/32} = \boxed{\frac{3}{4}}$.

Verified answers

$\frac{1}{4}; \frac{3}{4}; \text{No}; \frac{3}{64}; \frac{9}{64}; \frac{3}{32}; \frac{3}{4}$. After two like draws (red, red), the posterior stays $\frac{3}{4}$ *even when the third draw is blue* — the third student rationally ignores her own draw and a **cascade** forms.

Quiz Trap

Use the *urn-specific* signal probabilities $\frac{3}{4}/\frac{1}{4}$ (not the $\frac{2}{3}/\frac{1}{3}$ of the lecture’s 2-1 urn). Independence lets you *multiply* per-draw probabilities for a sequence. The denominator in Bayes is the *total* probability $\Pr[\text{evidence}]$, averaged over both urns.

Family 2 — Voting (Plurality & Borda)

Frequently Asked PYQ Concept

A preference schedule with vote counts; find the plurality winner, the Borda group ranking, and judge two statements about Borda.

6.1 Plurality

Example (sweets G, M, B), votes in thousands by first choice:

$$G : 44 + 14 + 20 = 78, \quad M : 70 + 22 = 92, \quad B : 80 + 39 = 119.$$

Plurality counts only first-place votes $\Rightarrow$ B wins (largest first-place tally, 119k).

6.2 Borda count

Example, 3 voters, 4 alternatives, points 3, 2, 1, 0:

$$1 : B \succ C \succ D \succ A, \quad 2 : C \succ D \succ A \succ B, \quad 3 : D \succ A \succ B \succ C.$$

$$A = 0+1+2 = 3, \quad B = 3+0+1 = 4, \quad C = 2+3+0 = 5, \quad D = 1+2+3 = 6 \Rightarrow D \succ C \succ B \succ A.$$

6.3 The two statements

I: “Borda always yields a complete, transitive ranking” — **true** (it sorts by a single score). **II:** “Borda’s outcome depends on the presence of seemingly irrelevant alternatives” — **true** (IIA failure). So $\boxed{\text{Both I and II are true.}}$ (A paper that words II as “does *not* depend” makes II *false* $\Rightarrow$ only I true — read II carefully.)

Quiz Trap

Plurality ignores all but first choices (so a majority-preferred option can still lose). Borda assigns $k - 1, \dots, 0$ — a common slip is to start from k or to count the last place as 1.

Family 3 — Gale–Shapley Matching

Frequently Asked PYQ Concept

A preference matrix; run Gale–Shapley with the stated proposing side and report each pairing.

7.1 Worked example (men propose; 4 men, 4 women)

From a Quiz-2 preference table, the lists are:

Men: $a : C \succ B \succ D \succ A$, $b : B \succ A \succ C \succ D$, $c : B \succ D \succ A \succ C$, $d : C \succ A \succ D \succ B$;

Women: $A : a \succ b \succ c \succ d$, $B : c \succ a \succ d \succ b$, $C : c \succ b \succ d \succ a$, $D : b \succ a \succ c \succ d$.

Run (men propose):

1. $a \rightarrow C$, $b \rightarrow B$, $c \rightarrow B$, $d \rightarrow C$. C has $\{a, d\}$, keeps d (rejects a); B has $\{b, c\}$, keeps c (rejects b).
2. $a \rightarrow B$ (next), $b \rightarrow A$ (next). B now $\{c, a\}$, keeps c (rejects a); A keeps b .
3. $a \rightarrow D$ (next). D keeps a . No rejections remain.

(A, b) , (B, c) , (C, d) , (D, a) i.e. $A-b$, $B-c$, $C-d$, $D-a$.

Verified answer

A matches b , B matches c , C matches d , D matches a . This is the *men-optimal* stable matching (men propose). *Unequal-size variant* (e.g. 4 men, 3 women with 0 entries marking unacceptable partners): run GS the same way; some proposer ends unmatched (one paper's answer is "Mr. $a \rightarrow \text{None}$ ").

Quiz Trap

Acceptances are *tentative*: a held man can be bumped later by a better proposer. Always let receivers re-compare their held partner against each new proposer. Men-proposing $\Rightarrow$ men-optimal/women-pessimal.

Family 4 — Repeated Games

Frequently Asked PYQ Concept

Two sub-types: (a) a *finitely* repeated Prisoner’s Dilemma; (b) an *infinitely* repeated game where cooperation needs a patient-enough discount factor.

8.1 Finitely repeated Prisoner’s Dilemma

Stage game $(C, C) = (2, 2)$, $(C, D) = (0, 3)$, $(D, C) = (3, 0)$, $(D, D) = (1, 1)$.

- **Unique stage NE** = (Defect, Defect) (Defect strictly dominates).
- **Finite repetition:** “In the *last* period, Defect is a dominant strategy *irrespective of history*” — the last round is a one-shot PD whatever happened before. By backward induction this unravels to “always defect.”

Verified answers

Unique NE = (Defect, Defect); in the last period *Defect is dominant irrespective of history*.

8.2 Infinitely repeated cooperation with discounting (the “campaign” game)

Two parties choose campaign levels; cooperating means $a_1 = a_2 = 0$. From the paper’s payoff function: cooperating yields each player a stage payoff of **0**; the *best one-shot deviation* is to campaign at level $\frac{1}{2}$, which yields the deviator $\frac{1}{4}$ that period; the punishment (stage Nash) payoff is -1 each.

Discount-factor condition via grim trigger

A one-shot deviation gains $T - R = \frac{1}{4} - 0 = \frac{1}{4}$ today, but triggers punishment $P = -1$ forever after (instead of the cooperative $R = 0$), losing $R - P = 0 - (-1) = 1$ each future period. Cooperation is sustainable (deviation unprofitable) iff

$$\underbrace{\frac{1}{4}}_{\text{gain}} \leq \underbrace{\frac{\delta}{1-\delta} \cdot 1}_{\text{discounted loss}} \iff \frac{1}{4}(1-\delta) \leq \delta \iff \frac{1}{4} \leq \frac{5}{4}\delta \iff \boxed{\delta \geq \frac{1}{5}}.$$

Hence the cooperative agreement is *not* supportable as a subgame-perfect equilibrium when $\boxed{\delta \leq \frac{1}{5}}$.

Verified answers

Cooperative stage payoff = 0; optimal deviation campaign level = $\frac{1}{2}$ (yielding $\frac{1}{4}$); cooperation sustained iff $\delta \geq \frac{1}{5}$, i.e. *not* supported iff $\delta \leq \frac{1}{5}$.

Quiz Trap

The threshold uses *three* numbers: the one-shot temptation gain, the per-period punishment loss, and the geometric factor $\frac{\delta}{1-\delta}$. Identify the stage-Nash (punishment) payoff correctly — here it is *negative* (-1), which is what makes deterrence work.

Family 5 — Bayesian Battle of Sexes

Frequently Asked PYQ Concept

Player 1 is unsure whether player 2 wants to *meet* (prob $\frac{1}{2}$) or *avoid* (prob $\frac{1}{2}$); fill the expected-payoff table a, b, c, d, p, q, r, t and find the Bayesian Nash equilibrium.

9.1 Topic & setup

Week 6 (Bayesian games). Type-matrices (player 1 chooses row B/S ; (u_1, u_2)):

	B	S		B	S
B	3, 1	0, 0	B	3, 0	0, 3
S	0, 0	1, 3	S	0, 1	1, 0
meet (prob $\frac{1}{2}$)			avoid (prob $\frac{1}{2}$)		

Player 2's strategy is (action if meet, action if avoid); player 1 averages over the two types.

9.2 Full working

Each cell is a $\frac{1}{2}/\frac{1}{2}$ average of player 1's payoff across the two matrices:

Player 1 \ Player 2	(B, B)	(B, S)	(S, B)	(S, S)
B	$a = 3$	$b = 1.5$	$c = 1.5$	$d = 0$
S	$p = 0$	$q = 0.5$	$r = 0.5$	$t = 1$

(e.g. b : player 1 = B , player 2 = (B, S) : meet gives 3, avoid gives 0, average 1.5.)

Bayesian NE. Meet-type of player 2 wants to *match* ($B \rightarrow B$, $S \rightarrow S$); avoid-type wants to *mismatch* ($B \rightarrow S$, $S \rightarrow B$). Anticipating this, player 1 compares: play $B \Rightarrow$ profile $(B, S) \Rightarrow$ payoff $b = 1.5$; play $S \Rightarrow$ profile $(S, B) \Rightarrow$ payoff $r = 0.5$. She picks B .

BNE: Player 1 plays B ; Player 2 plays $(B \text{ if meet, } S \text{ if avoid}) = (B, (B, S))$.

Verified answers

$a = 3$, $b = 1.5$, $c = 1.5$, $d = 0$, $p = 0$, $q = 0.5$, $r = 0.5$, $t = 1$; Bayesian NE = $(B, (B, S))$.

Quiz Trap

A “strategy” for the *informed* player (2) specifies an action *per type*, hence the four columns $(B, B), (B, S), (S, B), (S, S)$. Average over types *from player 1’s viewpoint*; then best-respond *type by type* for player 2 and once (in expectation) for player 1.

Family 6 — Evolutionarily Stable Strategies

Frequently Asked PYQ Concept

Two symmetric 2×2 games; find the Nash equilibria and which strategies are ESS.

10.1 The two games

		X	Y
Game 1	X	3, 3	0, 0
	Y	0, 0	0, 0

		X	Y
Game 2	X	3, 3	0, 0
	Y	0, 0	2, 2

10.2 Full working

Nash equilibria. Both are coordination games: (X, X) and (Y, Y) are pure NE in *each* (on the diagonal). (Verified answer: “ (X, X) and (Y, Y) ” for both games.)

ESS. Use the symmetric- 2×2 test: with own-payoffs a (vs same), c (mutant vs incumbent), a strategy is an ESS iff $a > c$, or ($a = c$ and it does strictly better against the mutant).

- **Game 1.** For X : own $a = 3$, mutant Y against X gets $c = 0$; $3 > 0 \Rightarrow X$ is ESS. For Y : own $d = 0$, invader X against Y gets 0 (tie), and against itself X gets $3 > 0$, so X invades $\Rightarrow Y$ is *not* ESS. ESS = X only.
- **Game 2.** For X : $3 > 0 \Rightarrow$ ESS. For Y : own $2 >$ invader X 's payoff 0 against $Y \Rightarrow$ ESS. ESS = X and Y .

Verified answers

Both games: pure NE = (X, X) and (Y, Y) . **Game 1** ESS = X only (the (Y, Y) equilibrium is *weak* — a mutant ties then wins, so Y is invadable). **Game 2** ESS = both X

and Y (both diagonal equilibria are *strict*).

Quiz Trap

A *strict* symmetric Nash equilibrium is always an ESS; a *weak* one (Game 1's (Y, Y) , where the off-diagonal payoff equals the diagonal) usually is *not*. ESS is the stricter concept: NE checks “no profitable deviation,” ESS additionally checks “resists drifting mutants.”

Family 7 — Design-Thinking / KDT Scenario Items

Scope note — please read

The 2025 T3 paper includes two scenario MCQs (“TravelEase” and “FoodFleet”) that test a **design-thinking / “KDT” process**, *not* game theory. This material does *not* appear in the Weeks 5–8 lecture slides — it seems to come from a separate strategy/innovation module bundled into the same exam sitting. The answers below are justified using standard, well-established design-thinking principles; the course-specific acronyms (“KDT”, “ENV model”) should be confirmed with the instructor. *Please verify these with the course instructor.*

11.1 Background: the design-thinking pipeline

The five-phase flow

Design thinking proceeds **Empathize → Define → Ideate → Prototype → Test** (a “double-diamond” of diverging then converging twice).

- **Empathize:** build *customer journey maps* (CJMs), personas; mark *emotional valleys* (pain points / friction).
- **Define:** *synthesise* — cluster the pain points into *problem themes*, then state the desired result.
- **Ideate:** diverge — generate many ideas (e.g. *silent brainstorming*) without early judgement.
- **Prototype & Test:** build at *appropriate fidelity* and *test with real customers* before committing.

11.2 The two questions

TravelEase

Two distinct ideas (AI trip-planning vs. social trip-sharing) both resonate with different customer segments; what is the sensible next KDT step? Picking by hype (“AI is trending”/“viral potential”) or merging both into one release prematurely all skip the evidence step. The disciplined move under design thinking is to **prototype both concepts at appropriate fidelity and test them with customers** — let user evidence, not opinion, decide.

Verified answer: Prototype both concepts at appropriate fidelity and test with customers.

FoodFleet

A CJM reveals “emotional valleys” around order delays and unclear driver updates; what is the next KDT step? The valleys are raw *Empathize*-phase findings; before ideating or defining results you must *synthesise* them. The correct next step is to **cluster the identified valleys into problem themes** (Define phase). Jumping to ideation (“silent brainstorming”), to the results model, or simplifying the map to hide the valleys would all skip the synthesis that makes ideation productive.

Verified answer: Cluster the identified valleys into problem themes.

★ Exam Tip

Pattern for these items: identify *which phase* the scenario is in, then pick the option that is the *disciplined next phase* — never the option that skips ahead on hype, merges prematurely, or hides the data. “Prototype & test rather than bet” and “synthesise pain points before ideating” are the two recurring morals.

Part III

Concepts Found Only in the PYQs

Gap-Filling Concepts

Simple Intuition

A few tools are *used* by Quiz 2 but only lightly *stated* in the slides. Here they are, each with just enough theory to solve the matching question cold.

12.1 Bayes' Rule, Operationally

The three forms you actually use

$$\Pr[A \mid B] = \frac{\Pr[A \cap B]}{\Pr[B]} = \frac{\Pr[B \mid A] \Pr[A]}{\Pr[B]}, \quad \Pr[B] = \sum_{\text{states } A_i} \Pr[B \mid A_i] \Pr[A_i].$$

Cascade recipe. (1) Write priors $\Pr[\text{maj-blue}] = \Pr[\text{maj-red}] = \frac{1}{2}$. (2) Write signal likelihoods (for the 3-1 urn, $\frac{3}{4}/\frac{1}{4}$). (3) For a *sequence*, multiply per-draw likelihoods (independence). (4) Total probability = average over both urns. (5) Posterior = (prior $\times$ likelihood)/total. A *cascade* has formed once the posterior — hence the optimal guess — no longer changes with the next private draw.

12.2 Expected Payoffs in Bayesian Games

Definition

The *informed* player's strategy lists an action *per type*; the *uninformed* player cannot see the type and so maximises the **type-averaged** payoff

$$U_1(a_1, \sigma_2) = \sum_{\text{types } \theta} \Pr[\theta] u_1(a_1, \sigma_2(\theta); \theta).$$

A Bayesian NE has every *type* of the informed player best-responding, and the uninformed player best-responding in expectation.

★ Exam Tip

Build the uninformed player's expected-payoff table first (one column per type-contingent strategy of the opponent), then read off best responses. This is exactly the a, b, c, d, p, q, r, t table.

12.3 Discounting, Geometric Series, and the One-Shot-Deviation Principle

Formula Insight

$$\sum_{t=0}^{\infty} \delta^t u = \frac{u}{1-\delta}, \quad \sum_{t=1}^{\infty} \delta^t u = \frac{\delta u}{1-\delta}.$$

One-shot-deviation principle: a strategy is subgame perfect iff *no* single-period deviation (then revert to the strategy) is profitable. Cooperation under grim trigger survives iff

$$\underbrace{(T-R)}_{\text{one-shot gain}} \leq \frac{\delta}{1-\delta} \underbrace{(R-P)}_{\text{per-period loss}} \iff \delta \geq \frac{T-R}{T-P}.$$

Reading the punishment payoff P . P is the *stage-game Nash* payoff that players revert to. In the campaign game $P = -1$ (mutual campaigning is costly), which is why $\delta \geq \frac{1}{5}$ suffices despite the cooperative payoff being only 0.

12.4 ESS vs. Strict/Weak Nash

Remember This

For a symmetric 2×2 game with own-payoffs a (vs same), b (vs the other), c (other vs you), d (other vs other):

$$(S, S) \text{ is NE} \iff a \geq c; \quad S \text{ is ESS} \iff a > c \text{ or } (a = c \text{ and } b > d).$$

Strict NE $\Rightarrow$ ESS. **Weak NE** (a tie, $a = c$) is usually *not* an ESS, because a neutral mutant can drift in and then out-compete. Every ESS is a symmetric Nash equilibrium; the converse fails.

12.5 Voting Pathologies Quoted in Questions

Remember This

Plurality counts only first places (ignores the rest). **Borda** ($k-1, \dots, 0$) always gives a complete, transitive ranking *but* violates **IIA** — an “irrelevant” option can flip two others — and invites **strategic misreporting**. **Condorcet paradox:** pairwise majority can cycle. **Arrow:** for ≥ 3 alternatives, Unanimity + IIA $\Rightarrow$ dictatorship.

Part IV

Quick Revision Sheets

Last-Minute Revision

13.1 One-Page Summary (Weeks 5–8)

The whole block on one screen

Repeated games: strategies depend on history. *Finite* + unique stage NE $\Rightarrow$ play stage NE every round (unravelling); *infinite* + patience sustains cooperation. **Grim trigger** works iff $\delta \geq \frac{T-R}{T-P}$.

Evolution: fitness = payoff; **ESS** = uninvadable strategy; $a > c$ (or $a = c, b > d$); Hawk–Dove ESS $p^* = V/C$.

Bayesian games: private *types* + prior; uninformed player averages over types; informed player best-responds per type. First-price auction (uniform): bid $\frac{n-1}{n}v$.

Cooperative games: (N, v) ; $MC_i = v(N) - v(N \setminus i)$; *core* = efficient with $\sum_S x_i \geq v(S)$.

Matching: stable = no blocking pair; Gale–Shapley always finds one (two-sided); proposer-optimal.

Cascades: sequential + observed actions $\Rightarrow$ copy the crowd (Bayes). **Voting:** plurality/Borda/Condorcet pathologies; **Arrow:** Unanimity + IIA + Non-dictatorship impossible (≥ 3).

13.2 Formula Card

Formula Insight

Discounting: $\sum_{t \geq 0} \delta^t u = \frac{u}{1-\delta}$; cooperate iff $\delta \geq \frac{T-R}{T-P}$.

ESS (2×2): $a > c$, or $(a = c \text{ and } b > d)$; Hawk–Dove $p^* = \frac{V}{C}$.

Bayes: $\Pr[A | B] = \frac{\Pr[B|A]\Pr[A]}{\Pr[B]}$; 3-1 urn signal = $\frac{3}{4}/\frac{1}{4}$.

Auction: $b(v) = \frac{n-1}{n}v$; $MC_i = v(N) - v(N \setminus \{i\})$.

Core: $\sum_N x_i = v(N)$, $\sum_S x_i \geq v(S) \forall S$.

Borda: points $k-1, k-2, \dots, 0$.

13.3 Algorithm Cheat-Sheet

★ Exam Tip

Cascade: priors $\rightarrow$ likelihoods $\rightarrow$ multiply for sequences $\rightarrow$ total prob $\rightarrow$ posterior; cascade = guess independent of own draw.

Repeated (finite): unique stage NE $\Rightarrow$ that NE every round.

Repeated (infinite): find $T, R, P \rightarrow \delta \geq \frac{T-R}{T-P}$.

ESS: compare own-vs-same against mutant-vs-you (a vs c); strict $\Rightarrow$ ESS.

Bayesian game: build expected-payoff table (average over types) $\rightarrow$ best-respond per type.

Gale–Shapley: proposers propose down lists; receivers keep best, reject rest; repeat.

Voting: plurality = first places; Borda = weighted sum $k-1 \dots 0$; check IIA for “irrelevant” flips.

13.4 Common Exam Traps

Quiz Trap

1. Using $\frac{2}{3}/\frac{1}{3}$ (the 2-1 urn) when the question states the 3-1 urn ($\frac{3}{4}/\frac{1}{4}$).
2. Forgetting the punishment payoff P can be *negative* when computing the δ threshold.
3. In Bayesian games, averaging the *wrong* player's payoff, or forgetting the opponent has a per-*type* action.
4. Calling a *weak* symmetric Nash equilibrium an ESS (it usually isn't).
5. Borda: starting points at k instead of $k-1$, or claiming Borda is immune to irrelevant alternatives.
6. Gale–Shapley: treating acceptances as final instead of tentative; mixing up which side is proposer-optimal.
7. Plurality: declaring a majority-preferred option the winner when it has fewer *first* places.

13.5 Five-Minute Pre-Exam Notes

Remember This

- 3-1 urn: one off-colour draw $\Rightarrow$ posterior $\frac{3}{4}$ for that colour's majority; (r,r,b) $\Rightarrow$ still $\frac{3}{4}$ red $\Rightarrow$ cascade.
- Finite PD: last period Defect dominant *irrespective of history*.
- Campaign game: cooperate payoff 0, deviate $\frac{1}{4}$ (level $\frac{1}{2}$), sustain iff $\delta \geq \frac{1}{5}$.
- Bayesian BoS: $a, b, c, d, p, q, r, t = 3, 1.5, 1.5, 0, 0, 0.5, 0.5, 1$; BNE $(B, (B, S))$.
- ESS: strict diagonal NE are ESS; weak ones aren't (Game 1 $\Rightarrow X$ only; Game 2 $\Rightarrow X$ and Y).
- GS (men propose): men-optimal; sample answer $(A, b), (B, c), (C, d), (D, a)$.
- Plurality $\Rightarrow B$; Borda $\Rightarrow D \succ C \succ B \succ A$; both Borda statements true.

Part V

Exam Strategy Guide

How to Attack Quiz 2

14.1 The 60-Second Triage

Simple Intuition

Spend the first minute *classifying*. Quiz 2 is six recognisable families; the family fixes the recipe.

If you see...	...do this
“urn”, “majority-blue/red”, “draws”	Bayes: priors $\rightarrow$ likelihoods ($\frac{3}{4}/\frac{1}{4}$) $\rightarrow$ posterior; watch for the cascade.
“plurality / Borda / ranking”	First places (plurality) or weighted sum $k = 1 \dots 0$ (Borda); check IIA.
“stable matching / Gale–Shapley”	Run deferred acceptance for the stated proposing side.
“repeated... finite”	Last period = stage NE irrespective of history.
“repeated infinitely / binding agreement / δ ”	$\delta \geq \frac{T-R}{T-P}$; find T, R, P (mind negative P).
“incomplete information / types / BoS”	Average the uninformed player’s payoff over types; best-respond per type.
“ESS / evolutionarily stable”	Compare a vs c ; strict diagonal NE $\Rightarrow$ ESS.
“core / coalition / value $v(S)$ ”	Efficiency + all coalition inequalities $\rightarrow$ feasible range.

14.2 Per-Family Playbook

★ Cascades

Write the 2×2 table of likelihoods first. For sequences, *multiply*. The posterior for the leading colour after two same draws is $\frac{3}{4}$ (3-1 urn); from then on guesses are uninformative — that *is* the cascade. Don’t recompute from scratch each part; reuse the running likelihoods.

★ Repeated games

Finite horizon: answer “stage NE” instantly. Infinite: the only arithmetic is $\delta \geq \frac{T-R}{T-P}$. Pull T (best deviation payoff), R (cooperation), P (stage-Nash punishment) straight from the matrix; “not supported” is the complementary inequality.

★ Bayesian games

Two moves: (1) collapse to the uninformed player’s expected-payoff table by averaging over types; (2) best-respond *type by type* for the informed player, then once for the uninformed player. The BoS template ($a=3, \dots, t=1$, BNE ($B, (B, S)$)) recurs — recognise it.

★ ESS

Find pure NE by the underline method; then for each diagonal NE check strictness. Strict $\Rightarrow$ ESS; a tie on the off-diagonal $\Rightarrow$ not ESS. For Hawk–Dove-type mixing, set the two type fitnesses equal ($p^* = V/C$).

★ Matching

Track a small table: each receiver’s currently-held proposer. Process proposals in rounds; bump held partners when a better proposer arrives. Stop when no one is rejected. Verify by scanning for a blocking pair. Proposer side = optimal side.

★ Voting

Plurality: tally first places only. Borda: assign $k-1, \dots, 0$, sum, sort. If asked about “irrelevant alternatives” or strategic voting, the answer almost always invokes the *IIA failure*.

14.3 Time Management & High-Frequency Topics

Remember This

Bank the quick wins first: finite-PD answer, plurality/Borda, single-draw Bayes, ESS underline. Reserve time for the multi-part cascade chain and the Bayesian $a-t$ table (most sub-marks). **Ranked frequency:** (1) information cascade; (2) Gale–Shapley; (3) Bayesian BoS; (4) repeated games (finite & discounting); (5) ESS; (6) voting. Never leave a true/false blank — a verified fact settles it.

Final word

Six families, six recipes. Recognise the family in the first minute, apply the recipe, check the equilibrium/posterior condition, and report exactly what is asked. With Part I for understanding and Parts II–IV for drill, that is full marks on Quiz 2.

*End of Book 2 — Quiz 2 Preparation Guide.
Good luck. Made by Anmol Kansal with AI.*